SOFTWARE PROJECT REPORT | WINDOWS DESKTOP APPLICATION

Weight Tracker

An offline-first personal measurement log built with C# WPF, .NET 10 and SQLite

Project idea

Weight Tracker is a small local Windows application for recording dated weight measurements and understanding change over time. It combines a quick daily-entry form with a dashboard, measurement history, a trend chart and Monday-to-Sunday averages. Data stays in a local SQLite database. The portable Windows x64 package is self-contained, so the target PC does not need a separate .NET installation.

System requirements

Platform	Windows desktop; C# / .NET 10 WPF, Windows x64 single-file deployment.
Record keeping	One measurement per date; accept only finite positive weights, allow an optional note, and support editing and deletion.
Calculations	Store weights in kilograms; convert kg/lb for display; report weekly averages using only actual entries and calendar weeks running Monday through Sunday.
Views & settings	Dashboard summaries, date-range trend chart, searchable history, weekly table, light/dark appearance, color scheme, precision and date format.
Data workflows	Local SQLite persistence, backup/validated restore, CSV and JSON export, validated JSON import, and explicit record deletion.
Privacy & reliability	No account, telemetry, sync or network service is designed into the app. Preserve user data locally and report storage errors rather than silently moving it.

> **Scope:** This is a personal tracker, not a medical device or a cloud service. “Offline by design” describes the implementation and static source audit; runtime testing with Windows networking blocked remains outstanding.

Project evidence and implementation details summarized from the source, README, test harness and generated Windows package.

System architecture & modules

1 · WPF UI	2 · Domain logic	3 · SQLite store	4 · Local workflows	5 · Quality & release
Entry, dashboard, history, chart and settings	Units, dates, ranges and summaries	Records, settings, weekly cache and transactions	Quotes, CSV/JSON and backups	Core checks, source audit and publish script

The WPF window calls typed C# services. The store is the source of truth; weekly aggregates are recalculated within write transactions. The chart and summaries render locally from query results.

Module 1 — User interface and visualization

Approach & reasoning: Use a native WPF window and a few clear screens rather than add a web stack or chart dependency. Keep entry, summaries and trend visible together; use WPF shapes for the chart.

Workflow: Build the Overview form and summary cards, then connect navigation, searchable/editable history, weekly table, settings, date-range selector and chart to service methods.

Testing strategy: Launch the packaged app, inspect populated dashboard and weekly screens, and verify navigation and rendering with synthetic local data. Automated checks focus on the underlying services, not pixel-level UI.

AI tool: Copilot in VS Code supported edits against the local XAML and C# context; build and launch checks provided independent verification.

Representative prompt (summary): "Build a native WPF dashboard with entry, history, weekly averages, settings and a local trend chart."

Module 2 — Measurement rules and analytics

Approach & reasoning: Keep kilograms as the canonical value so display-unit changes do not rewrite stored records. Centralize conversions, Monday/Sunday boundaries, date ranges and change calculations in `WeightService`.

Workflow: Add positive finite-value validation and kg/lb conversion; derive date ranges and averages from actual dated entries; refresh summary and chart after add, edit or delete.

Testing strategy: Check pound conversion, week boundaries crossing a month, leap-day month ranges, correct record counts and recalculation after an edit or deletion.

AI tool: Copilot in VS Code helped draft and refine service logic; deterministic test assertions checked the result.

Representative prompt (summary): "Keep weights in kg internally, validate finite positive values, and compute Monday–Sunday averages from recorded entries only."

Development process by module

Module 3 — SQLite persistence

Approach & reasoning: Use a compact local database with parameterized commands and transactions. Enforce unique dates and valid weights in the schema; keep settings and weekly summaries local as well.

Workflow: Create schema version 1, implement record/settings queries, save and delete operations, and recalculate weekly aggregates in the same transaction as each mutation.

Testing strategy: Exercise save, duplicate-date rejection, edit/delete, persisted settings, weekly aggregate counts and SQLite backup/restore through isolated temporary databases.

AI tool: Copilot in VS Code was used to work within the existing C# service and project conventions; the test harness and SQLite execution verified behavior.

Representative prompt (summary): "Persist measurements and settings locally; make each change and its weekly-summary recalculation atomic."

Module 4 — Local resources and data exchange

Approach & reasoning: Bundle quotes as JSON and keep import/export self-contained. Validate an entire JSON import before writing so malformed input cannot partially replace existing entries; use SQLite's backup API for database copies.

Workflow: Load and validate local quote records, choose a daily quote deterministically, export kg-based CSV/JSON and import validated JSON with conflict updates.

Testing strategy: Verify bundled quote count and selection, JSON round-trip, invalid-import non-mutation, and database backup/restore.

AI tool: Copilot in VS Code assisted with the service implementation; filesystem and database assertions confirmed the file workflows.

Representative prompt (summary): "Validate all imported measurements before applying changes; malformed imports must leave the existing database unchanged."

Module 5 — Quality checks and packaging

Approach & reasoning: Keep verification runnable without a separate test framework; distribute a self-contained win-x64 executable with local resources and data-location guidance.

Workflow: Run the console service checks, scan C# source for common network-client APIs, then use the Windows script to publish the single-file app and copy its guides/resources.

Testing strategy: 14 core checks pass; the static source network audit reports no configured clients; startup/shutdown smoke test passed. The audit is not proof of OS network isolation.

AI tool: Copilot in VS Code assisted with source and test changes; .NET SDK, test runner and PowerShell audit were used for reproducible checks.

Representative prompt (summary): "Add focused tests for date boundaries, editing, import safety and backup; keep the release self-contained for Windows x64."

Challenges, tool comparison & next steps

Biggest challenge — date-correct summaries. Week boundaries must remain Monday–Sunday across month and year changes, and averages must include only recorded measurements. An edit or delete must also update the corresponding aggregates. Focused edge-case tests helped pin down these rules.

Data integrity and portability. Import validation needs to finish before database mutation; backup restore needs an integrity/schema check. For portable deployment, the user database must remain beside the app rather than inside .NET's temporary native-library extraction directory.

Offline claims. A source scan can catch common networking APIs, but cannot prove that a binary or operating system will never communicate. Accordingly, the report does not claim that firewall-blocked runtime behavior has been verified.

Tool comparison and choice

Copilot (through the Copilot SDK in VS Code) was the AI coding assistant used. Its main advantage for this project was staying close to the open workspace: it could help refine changes across WPF, services, tests and documentation in the same repository. It was preferred over a standalone chat workflow for that integrated code context and edit/test loop.

Tool / role	Contribution and limits
Copilot in VS Code	AI assistance for implementation and refinement in project context. Suggestions still required review and executable tests.
.NET SDK / test harness	Repeatable builds and 14 concrete service checks. Not an AI assistant; supplied evidence beyond generated code.
PowerShell source audit	Repeatable scan for common network-related source patterns. Not a binary inspection or firewall test.
Other AI editors / agents	Not benchmarked on this project; no measured quality or speed comparison is claimed.

What I would improve next

- Add automated Windows UI/accessibility tests for navigation, validation, dialogs and chart rendering.
- Complete a controlled firewall-blocked runtime test and record the firewall/packet evidence.
- Add database migration tests and broader edge cases for empty, duplicate and boundary-date data.
- Review quote provenance and attribution before redistribution; maintain a verified source list.

> **Evidence snapshot:** 14 core checks passed; the static source audit passed; the app opened and closed cleanly. The screenshots on the next page use synthetic sample records in an isolated database, not the existing portable package database.

AI prompt examples in this report are concise representative summaries of the implementation requests, not a verbatim chat transcript.

Working system evidence

Captured from the running Windows WPF application using synthetic sample records and a separate local database. The dashboard shows recent change, weekly summaries and the local trend; the second view shows the Monday–Sunday aggregate table.

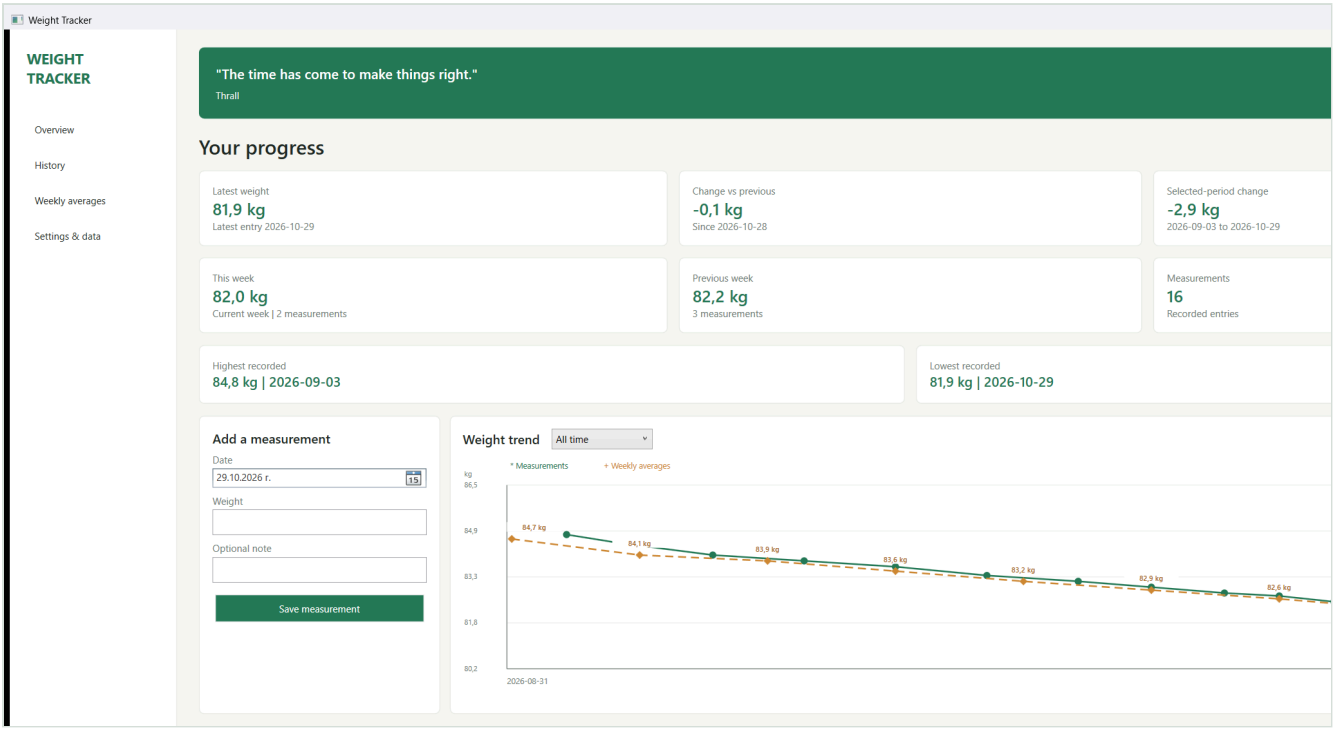


Figure 1. Overview dashboard with 12 synthetic entries, current/prior-week summaries and the local trend chart. Evidence captured 29 October 2026.

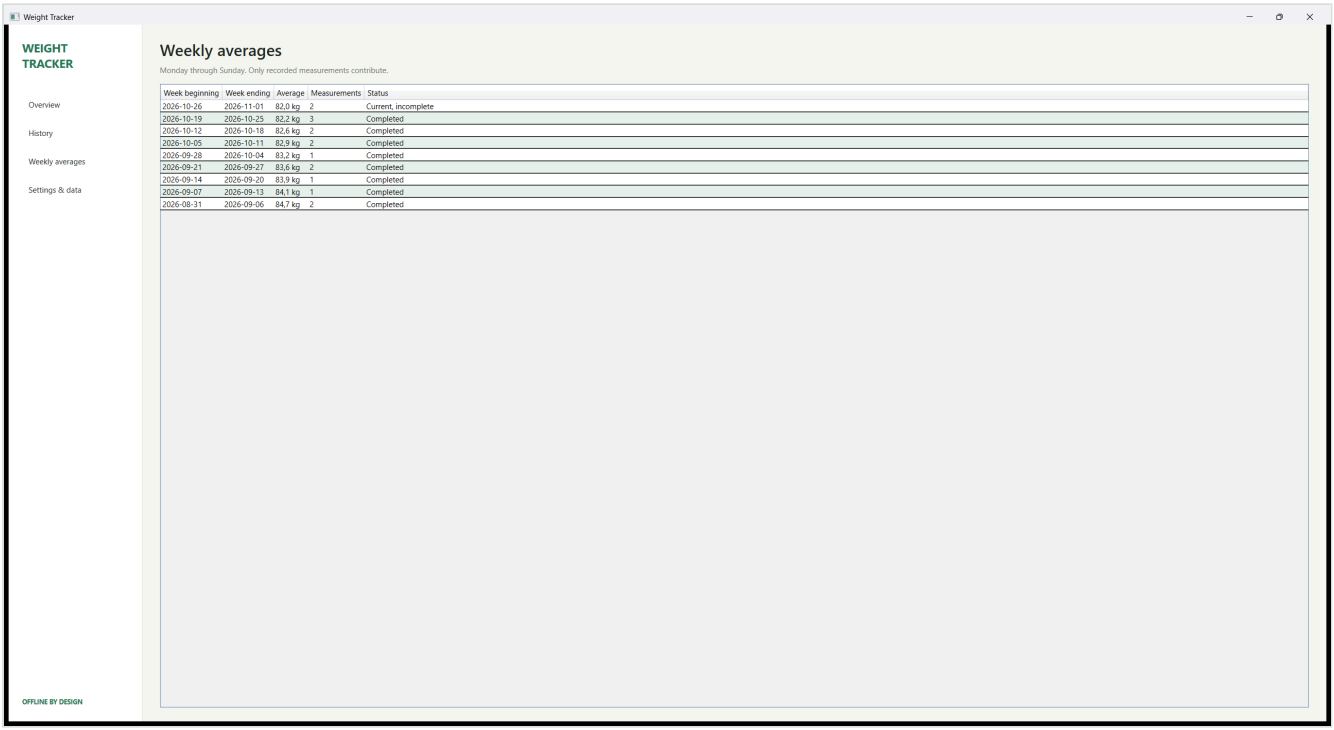


Figure 2. Weekly averages screen showing week start/end, average weight, measurement count and current/completed status. Evidence captured 29 October 2026.

The demonstration data is only for screenshots; it is not included in the published application or the user's portable database.